SAVEETHA SCHOOL OF ENGINEERING
C PROGRAMMING ASSIGNMENT – Grama Jyothi Billing Program

Field	Entry
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Register Number	1912260562
Course Code & Name	CSA0209 C PROGRAMMING
Slot	B
Assignment / Task Title	Design, Implement and Validate a Reliable Billing Program
GitHub Repository URL	https://github.com/bavadarini28 /CSA0209-1912260562-BAVADARANI.S.git
Submission Date	12 NOVEMBER 2026

TASK 5 – Header Files and Conditional Compilation

Aim

To organise constants and prototypes into a header file and add a DEBUG mode.

tariff.h

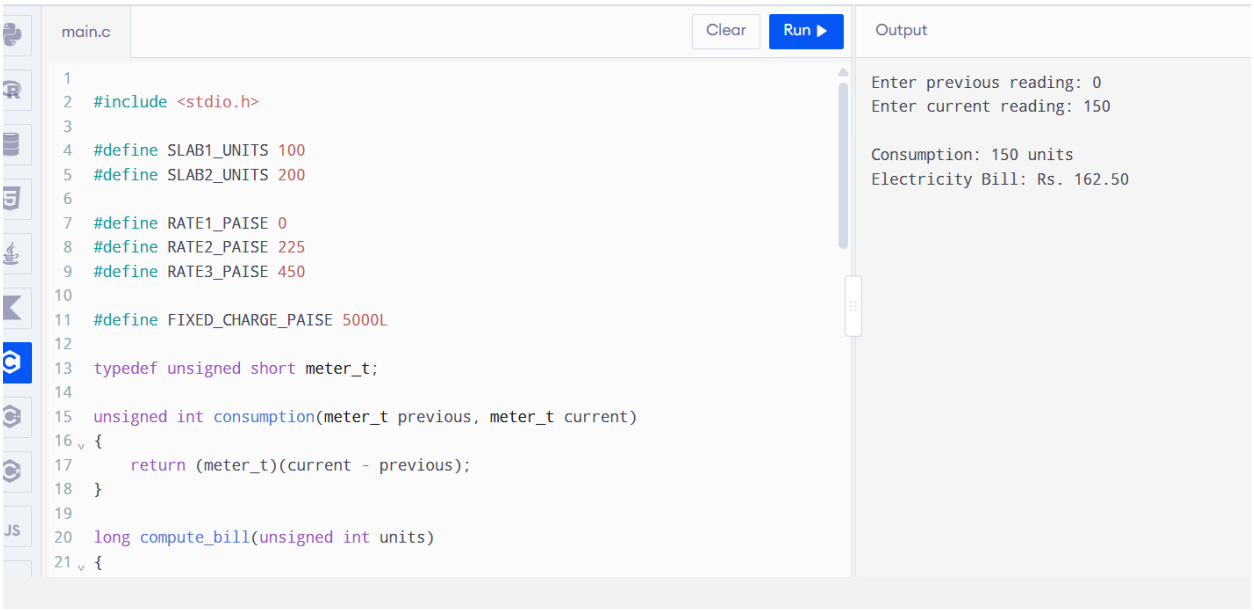
```
#ifndef TARIFF_H
#define TARIFF_H
#define SLAB1_UNITS 100
#define SLAB2_UNITS 200
#define RATE1_PAISE 0
#define RATE2_PAISE 225
#define RATE3_PAISE 450
#define FIXED_CHARGE_PAISE 5000L
#define RUPEES_TO_PAISE(r) ((long)(r) * 100L)
#define MAX(a,b) (((a) > (b)) ? (a) : (b))
#define SQUARE(x) ((x) * (x))
#define MAX_PLAUSIBLE_UNITS 5000u
typedef unsigned short meter_t;
unsigned int consumption(meter_t previous, meter_t current);
long compute_bill(unsigned int units);
#endif
```

Structure of the program: #include lines, then #define/typedef/global declarations (from the header), the helper functions consumption() and compute_bill(), and finally main(). Include guards (#ifndef TARIFF_H … #endif) stop the header from being processed twice, which would cause redefinition errors for typedefs and prototypes. #include <file> searches the system directories (standard library); #include "file" searches the current directory first, then the system ones.

Debug mode, inside compute_bill():

```
#ifdef DEBUG
   printf("[DEBUG] units=%u paise=%ld\n", units, paise);
#endif
$ gcc -DDEBUG -o billing_d billing.c && echo "1 0 150" | ./billing_d
[DEBUG] units=150 paise=16250
House 1: 150 units -> Rs 162.50
```

Programiz
C Online Compiler



Alternative Concept – run-time flag

Use int debug = 0; and if (debug) printf(...). Time: one extra comparison (a predictable branch) on every call, O(1) but non-zero – about 800,000 extra tests per district run, negligible in practice. Space: the debug code and format strings stay in the executable; with #ifdef they are removed from the release build. The flag has the advantage that debugging can be switched on without recompiling.

Result: Header with guards and a working DEBUG mode.